CSC418: Modeling and Simulation

Project Specification

B.Sc. Computer Science | University of Lagos

Course Code	CSC418
Course Title	Modeling and Simulation
Programme	B.Sc. Computer Science
Institution	University of Lagos
Submission Email	csc.class.projects@gmail.com
Submission Deadline	April 30, 2026
Online Project Defense	May 1, 2026

Task

Model and simulate the problem assigned to you and report using the guidelines below. Submit your report and simulation code by email to

csc.class.projects@gmail.com by the deadline stated below.

Guidelines

The following guidelines define the structure and content of your project report. Each section must be addressed clearly and thoroughly. A well-organised and detailed submission will attract higher marks.

1) Problem

State the problem assigned to you verbatim, exactly as it was given. Do not paraphrase or alter the wording. Clearly identify the system being modelled, the key entities involved, and the performance metrics or outputs of interest.

2) Model

Describe the model of the system (problem) you are simulating. Your description must include:

- A schematic diagram clearly illustrating the structure and components of the system model.

- Identification of the entities, resources, queues, and flow of events in the system.
- Assumptions made in formulating the model, including any simplifications applied to the real-world system.

3) SALT Framework

Give a structured overview of the Simulator of ALL Things (SALT) framework, which serves as the foundation for your simulation development. Your response should include:

i) Overview: Provide a concise description of the SALT framework, its purpose, architecture, and how it supports the development of discrete event simulations. Kindly refer to the Zoom recordings on the purpose and development of SALT. If any recordings are missing, kindly request them.

ii) Static Class Diagram: Present a static class diagram (UML) of the SALT framework. The diagram should clearly show the key classes, their attributes, methods, and relationships (inheritance, composition, associations).

iii) Object-Event Graph (OEG) — OODES Explanation: Explain, using an Object-Event Graph (OEG) diagram, how the SALT framework functions as an Object-Oriented Discrete Event Simulator (OODES). Specifically, demonstrate how events are mapped to object handlers using the OEG. Your explanation should clearly identify the object plane and the event plane, and describe how events trigger transitions and invoke handler methods on simulation objects.

4) Simulation Program Organization and Logic

This section should present a comprehensive description of how your simulation program is organised and how its logic operates. The following sub-sections are required:

i) OEG Model of the Problem: Present a detailed Object-Event Graph (OEG) model specifically for your assigned problem, using a clearly drawn OEG diagram. Your explanation must cover:

- The Object Plane: Identify and describe all simulation objects (nodes) and their roles within the system.
- The Event Plane: Identify and describe all events and their sequencing logic.
- Edges: Label and explain every edge connecting nodes in both planes. A clear and detailed explanation of all labelled edges will attract additional marks.

ii) Event Handling Flow Charts: Describe the logic of the simulator using event handling flow charts. Each significant event type should have its own flow chart showing the sequence of actions, decisions, and state changes triggered when that event is processed.

iii) Statistics Computation: Describe, using mathematical equations, how you intend to compute the statistics required for your simulation problem. For each statistic, provide:

- The name and definition of the statistic.
- The formula or equation used to compute it.
- How the data needed to compute the statistic is collected during the simulation.

5) Simulation Output and Discussion

Run multiple simulations of your model (using different input parameter configurations) and present the results. Your submission for this section should include:

i) Simulation Output Table: Generate the output of multiple simulations showing, for each run, only a summary of the input parameters and the resulting output statistics. Present this as a clearly labelled table.

ii) Discussion: Discuss the output of your simulations. Explain the observed trends, patterns, or anomalies across runs. Relate the output to the behaviour of the real-world system being modelled.

iii) Optimization: Explain how the problem can be optimized based on your simulation results. Identify which input parameters most significantly affect performance and suggest practical strategies for improving system efficiency or achieving desired outcomes.

6) Appendix — Simulation Source Code (C#)

In your appendix, list the complete source code of your simulator written in C#. Adhere strictly to the following rules:

- Include only your simulation code. Do not include any SALT library code in the appendix.
- Do not modify any code in the SALT library directly. All customization must be achieved through inheritance — you may subclass SALT classes to extend or override their behavior.
- Ensure your code is well-formatted, properly commented, and logically structured.
- Each class and significant method should be preceded by a brief comment explaining its purpose.

> **Note:** Your implementation should not modify any code in the SALT library directly. You may use inheritance to modify or extend classes in SALT.

Submission Details

Your complete project report (PDF or Word document) and simulation source code (.cs files or zipped project folder) must be submitted as email attachments to the address below.

Submission Email	csc.class.projects@gmail.com
Attachments Required	Project Report (PDF/DOCX) + Simulation Code (C# / ZIP)
Email Subject Format	[CSC418] – [Student Name] – [Matric Number]

Submission Deadline	**Online Project Defense**
April 30, 2026	**May 1, 2026**
Email to csc.class.projects@gmail.com	Online (details to be communicated)

Assessment Criteria

Your project will be assessed based on the quality and completeness of each section. The following table provides a general guide to the marking distribution:

Section	Max Marks	Remarks
1. Problem Statement	**5**	Accuracy and completeness
2. System Model (Schematic)	**10**	Clarity and correctness
3. SALT Framework (Class Diagram & OEG)	**20**	Depth of understanding
4i. OEG Model of Problem	**20**	Detail and explanation of edges
4ii. Event Handling Flow Charts	**15**	Logic and completeness
4iii. Statistics Equations	**10**	Correctness of formulae
5. Simulation Output & Discussion	**10**	Quality of analysis
6. Appendix — C# Source Code	**10**	Code quality and structure
TOTAL	**100**	

Good luck!